

SREDNY VASYUGAN, Russian Federation

WMO: 291110

Lat: 59.217N

Lon: 78.233E

Elev: 226

StdP: 14.58

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.0	(c) -32.8	(d) -43.3	(e) 0.5	(f) -38.2	(g) -37.3	(h) 0.7	(i) -32.4	(j) 15.2	(k) 15.2	(l) 13.8	(m) 11.3	(n) 1.6	(o) 50	(p) 0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	83.7	66.0	80.8	65.3	77.5	64.0	69.0	78.9	67.2	76.8	65.5	74.6	5.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	94.9	74.0	63.7	89.0	71.9	61.9	83.4	70.1	33.4	79.0	31.9	76.9	30.5	74.7	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	13.3	11.8	-41.8	88.7	7.8	2.7	-47.4	90.6	-52.0	92.2	-56.3	93.7	-61.9	95.6
			-42.3	72.2	7.1	2.5	-47.4	74.0	-51.6	75.5	-55.6	76.9	-60.8	78.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	31.8	-3.9	2.4	20.1	33.5	47.2	61.2	65.0	58.7	47.4	33.7	13.1	1.6	(6)	
(7)		DBStd	26.52	16.31	14.78	11.95	11.11	10.50	9.27	6.68	6.91	8.25	9.52	15.96	16.62	(7)	
(8)		HDD50	7904	1672	1332	927	500	183	19	0	8	146	509	1107	1501	(8)	
(9)		HDD65	12287	2137	1752	1392	945	557	178	85	217	531	970	1557	1966	(9)	
(10)		CDD50	1273	0	0	0	5	97	355	466	279	67	4	0	0	(10)	
(11)		CDD65	182	0	0	0	0	7	64	86	23	2	0	0	0	(11)	
(12)		CDH74	1518	0	0	0	0	81	547	688	192	10	0	0	0	(12)	
(13)		CDH80	342	0	0	0	0	19	140	153	30	0	0	0	0	(13)	
(14)	Wind	WSAvg	5.4	5.0	5.4	6.3	6.3	6.3	4.9	4.2	4.4	5.2	5.8	5.9	5.5	(14)	
(15)	Precipitation	PrecAvg	21.7	1.1	0.8	0.8	1.3	2.1	2.9	2.8	3.2	2.0	1.9	1.6	1.3	(15)	
(16)		PrecMax	29.4	1.9	2.7	2.0	3.1	5.1	7.1	7.3	6.1	3.9	4.3	3.1	2.4	(16)	
(17)		PrecMin	15.2	0.3	0.2	0.1	0.3	0.6	0.2	0.3	0.0	0.5	0.7	0.6	0.2	(17)	
(18)		PrecStd	3.8	0.4	0.5	0.4	0.7	1.0	1.4	1.5	1.6	0.9	0.8	0.6	0.5	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	33.3	46.9	66.3	82.9	87.9	87.8	83.5	74.5	59.6	40.4	32.9	(19)	
(20)			MCWB	30.9	31.0	38.0	50.8	61.7	65.4	69.8	67.1	58.8	50.1	38.2	31.9	(20)	
(21)		2%	DB	27.1	30.0	41.5	58.7	75.4	84.0	83.7	79.2	69.3	54.3	36.7	30.0	(21)	
(22)			MCWB	25.8	28.2	34.2	46.7	57.4	65.2	67.0	65.4	58.1	46.8	35.1	28.9	(22)	
(23)		5%	DB	22.4	26.5	37.8	53.8	70.2	80.3	81.2	75.2	64.8	50.7	34.3	27.4	(23)	
(24)			MCWB	21.3	24.8	32.6	43.8	55.2	64.2	66.5	63.7	55.8	45.4	33.0	26.2	(24)	
(25)		10%	DB	18.2	22.3	35.0	49.4	64.7	76.6	78.0	71.1	60.4	46.8	32.4	23.7	(25)	
(26)			MCWB	17.2	20.4	31.2	41.2	52.7	63.1	65.2	62.1	53.7	42.9	31.2	22.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.1	31.8	38.3	52.1	62.9	70.6	73.0	69.6	62.2	51.3	38.3	32.0	(27)	
(28)			MCDWB	32.2	32.9	45.6	65.0	80.0	82.1	84.1	79.7	70.0	56.3	39.7	32.5	(28)	
(29)		2%	WB	25.8	28.4	35.2	48.1	59.2	67.8	70.1	66.8	59.1	48.5	35.3	29.0	(29)	
(30)			MCDWB	27.0	29.8	39.3	57.8	71.8	78.8	80.0	75.9	66.9	53.0	36.4	29.9	(30)	
(31)		5%	WB	21.4	24.7	33.3	45.0	56.4	66.0	68.1	64.9	56.6	45.8	33.2	26.3	(31)	
(32)			MCDWB	22.3	26.3	37.0	52.1	67.5	76.7	77.1	72.7	63.7	50.0	34.1	27.3	(32)	
(33)		10%	WB	17.2	20.5	31.5	41.8	53.4	64.0	66.4	62.9	54.1	42.9	31.2	22.7	(33)	
(34)			MCDWB	18.2	22.2	35.2	48.7	63.6	73.8	75.4	70.2	59.8	46.6	32.4	23.8	(34)	
(35)	Mean Daily Temperature Range			MDBR	12.7	14.7	16.5	17.2	17.1	16.7	17.2	16.0	13.6	9.7	10.8	12.3	(35)
(36)		5% DB	MCDBR	11.8	11.4	16.9	22.0	25.1	22.5	22.6	23.0	21.1	14.6	9.3	12.8	(36)	
(37)			MCWBR	11.7	10.4	11.8	13.2	11.9	8.7	9.5	11.0	11.4	9.4	9.1	12.3	(37)	
(38)		5% WB	MCDBR	11.8	11.0	14.0	20.3	22.3	19.7	19.3	19.8	18.5	13.6	8.6	12.4	(38)	
(39)			MCWBR	11.6	10.2	10.7	13.1	12.0	9.0	9.6	10.6	10.7	9.4	8.6	12.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.269	0.307	0.359	0.385	0.390	0.407	0.446	0.413	0.365	0.335	0.287	0.267	(40)	
(41)		taud		1.996	2.015	1.951	2.070	2.289	2.319	2.186	2.264	2.357	2.269	2.056	1.938	(41)	
(42)		Ebn at Noon		156	213	236	256	266	262	247	246	241	207	147	108	(42)	
(43)		Edh at Noon		17	28	41	43	37	37	41	36	28	22	16	12	(43)	
(44)	All-Sky Solar Radiation	RadAvg		172	474	941	1383	1546	1760	1608	1187	757	358	157	95	(44)	
(45)		RadStd		17	36	68	132	164	161	166	131	98	31	34	10	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page